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The Pharmacologic Management of ADHD

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Please email to set up a meeting

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Session Objectives

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1. Explain the mechanism(s) of action for ADHD medications.
2. Describe the clinical benefits of ADHD drug therapy.
3. Recognize key differences between medications used to treat ADHD.
4. Identify adverse effects related to ADHD pharmacotherapy.
5. Recommend appropriate monitoring parameters for patients on ADHD medications.

Source: Course Syllabus

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1. COMLEX-USA
    1. Dimension 2: Clinical Presentations

        Human Development, Reproduction, and Sexuality

2. USMLE® Content

    IV. Behavioral Health

    7. Disorders originating in infancy/childhood

        ii. Attention-deficit/hyperactivity disorder

ADHD Management

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- Pharmacotherapy is only one modality
- Psychosocial treatments
  - Behavioral therapy
  - Training interventions
  - Mindfulness, cognitive training, EEG biofeedback, counseling, dietary modification

AAP Clinical Practice Guidelines. Pediatrics. 2019; 144(4):e2019528.

ADHD Management

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- Key considerations regarding pharmacologic treatment:
  - The goal is to control symptoms and behavior that impair learning and development.
  - Individualized and specific behavioral, academic, and social target goals and treatments should be identified.
  - Response to medications is not a paradoxical effect.

DePiro JT, 8e, 2011 (Ch. 72) & DePiro JT, 10e, 2017 (Ch. 63)

Default Mode Network

- Default attention mode is established via a network between the prefrontal cortex, the medial parietal lobe (precuneus), and the posterior cingulate
  - Normally, this default is active during a resting state and inhibited during active attention
- A lack of connectivity within this network in ADHD may result in the default mode not being suppressed
  - Consequence is inattentiveness and lack of impulse control

Friedman LA , Rapoport JL. Brain development in ADHD. Curr Opin Neurobiol. 2015 Feb;30:106-11.

Neurobiology of ADHD & Drug Response

- Proposed benefit of medications
- Increase in dopamine (DA) reduces background ‘noise’ (restore inhibitory activity of dopamine)
- Increase in norepinephrine (Nor) – may strengthen connections (increase signals)

Purper-Ouakil D, Ramoz N, Lepagnol-Bestel AM, et al. Neurobiology of attention deficit/hyperactivity disorder. Pediatr Res. 2011 May; 69(5 Pt 2):69R-76R.

Biosynthesis of Catecholamines

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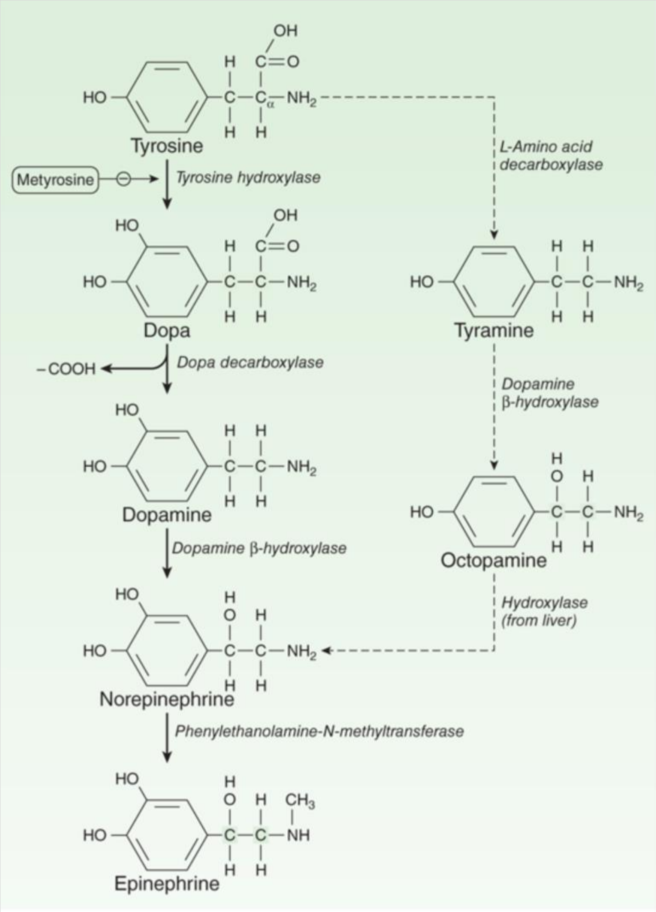
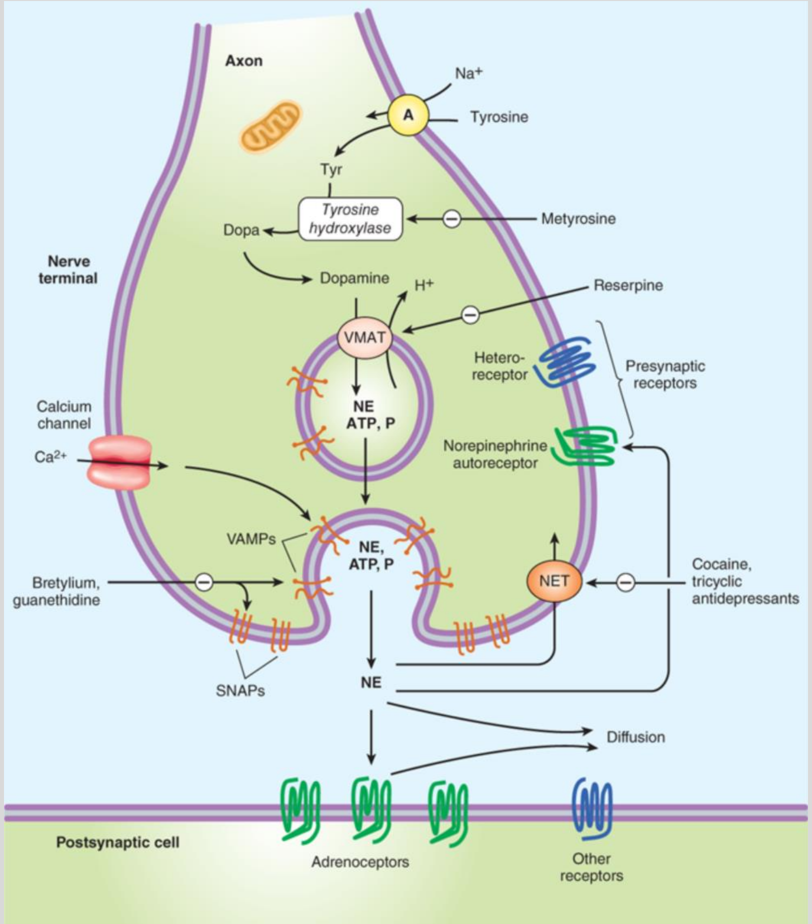


Katzung's Basic and Clin Pharm, 16e, 2024; Ch. 6.(Fig. 6-5).

Generalized Noradrenergic Junction



Katzung's Basic and Clin Pharm, 16e, 2024;
Ch. 6. (Fig. 6-4)

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(Relative) Classification of Adrenergic Agonists

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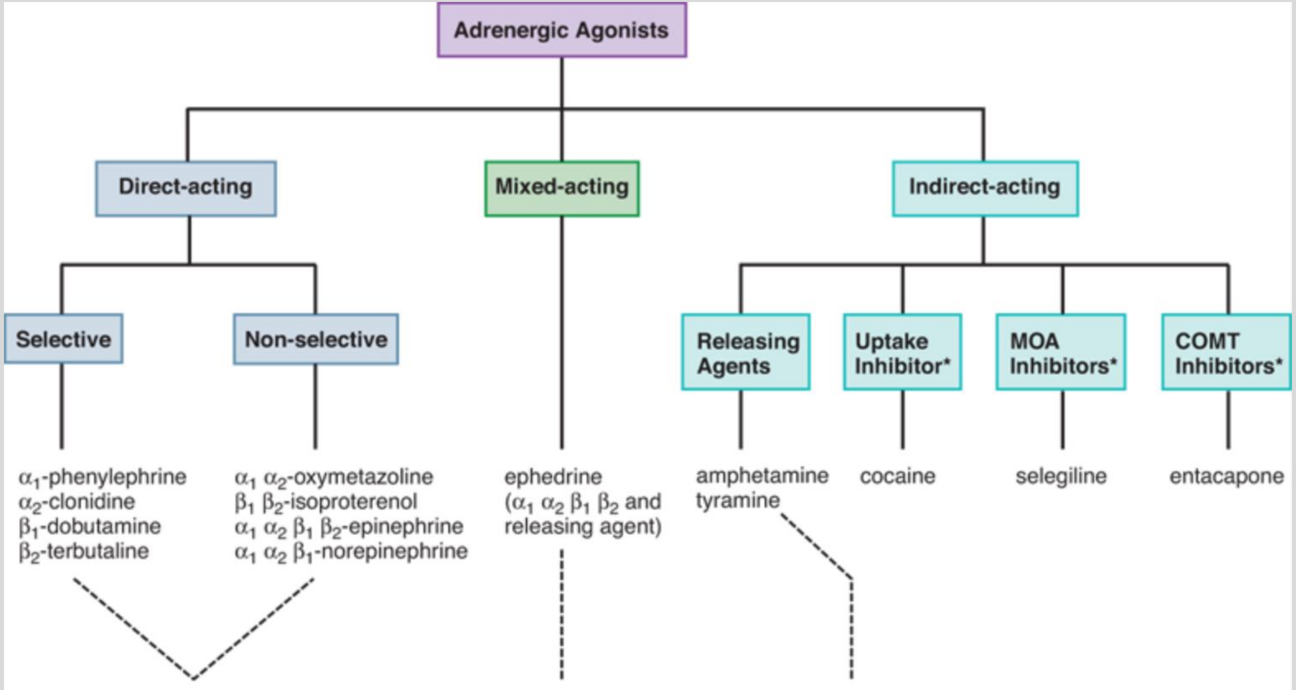

Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, 2023.
Ch. 14. Adrenergic Agonists and Antagonists (Fig. 14-1)

Pharmacologic Targeting of Monoamine Transporters

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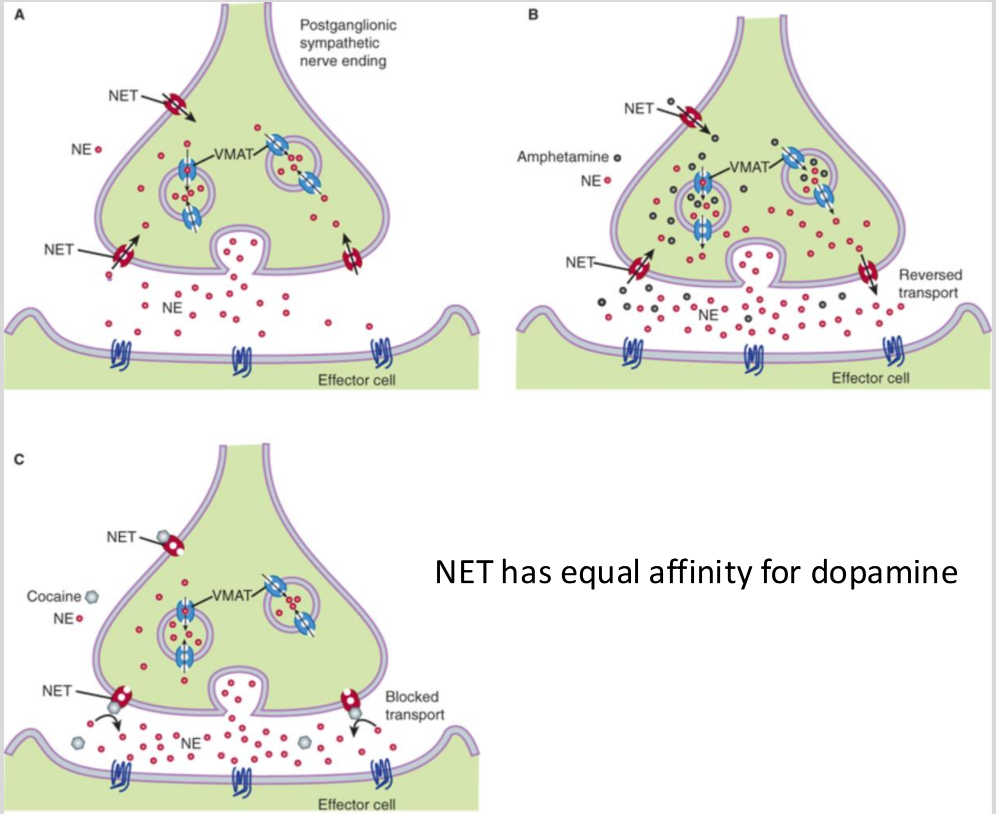

NET has equal affinity for dopamine

ADHD Pharmacotherapy Timeline

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1937 - Benzedrine (racemic amphetamine)
1943 - Desoxyn (methamphetamine hydrochloride)
1955 - Ritalin (methylphenidate)
1955-1983 - Biphetamine (mixed amphetamine/dextroamphetamine)
1960 - Adderall (mixed amphetamine /dextroamphetamine salts)
1975-2003 - Cylert (pemoline)
1976 - Dextrostat (dextroamphetamine)
1976 - Dexedrine (dextroamphetamine)
1982 - Ritalin SR
1999 - Metadate ER (methylphenidate)
2000 - Concerta (methylphenidate)
2000 - Methylin ER (methylphenidate)
2001 - Metadate CD (methylphenidate)
2001 - Focalin (dexmethylphenidate)
2001 - Adderall XR
2002 - Ritalin LA
2002 - Methylin (methylphenidate) oral solution and chewable tablet
2002 - Strattera (atomoxetine)
2005 - Focalin XR (dexmethylphenidate)
2006 - Daytrana (methylphenidate patch)
2007 - Vyvanse (lisdexamfetamine dimesylate)
2008 - Procentra (liquid dextroamphetamine)
2009 - Intuniv (guanfacine hydrochloride)
2010 - Kapvay (clonidine hydrocloride)
2012 - Quillivant XR (liquid methylphenidate)
2016 - Adzenys XR-ODT (amphetamine oral disintegrating tablet)
2016 - Quillichew ER (chewable methylphenidate)
2017 – Mydayis
2019 – Jornay PM
2021 – Azstarys, Qelbree

https://www.verywellmind.com/adhd-history-of-adhd-2633127; Accessed 1/9/26

Case

- TJ is an 8-year-old boy who is in the 3rd grade. He was recently diagnosed with ADHD after his teacher noticed that he does not follow through on tasks, is easily distracted, is very forgetful, and does not seem to listen.
- Social and behavioral interventions have been initiated at home and school. However, the decision has been made to begin pharmacotherapy since TJ continues to struggle in the classroom.
- What is a reasonable therapy to recommend?

Stimulants

- Stimulants are first-line therapy, unless the patient has certain co-morbidities.
- The stimulant group includes amphetamines & methylphenidate.
    - Approximately 40% respond to both.
    - 40% respond to only one.
    - ADHD subtype does not predict response to specific agent.

AAP Clinical Practice Guidelines. Pediatrics. 2019; 144(4):e2019528.

Stimulants

- Several formulations are available
  - Immediate-release
    - Can be dosed in lower amounts for younger children
    - Must be dosed 2 to 3 times a day
  - Delayed-release
    - Some may be dosed once daily, preventing the need for a dose during school
  - IR and delayed-release combination products
    - Capsule-filled beads
  - Transdermal delivery
  - Oral suspension

Stimulants: Amphetamines

- MOA: Increase DA and Nor in synaptic cleft via:
  displacement from pre-synaptic storage vesicle;
  reverse transport; blockade of reuptake transporters (non-selective)
- Isomers of amphetamine
    - *d*-isomer (dextroamphetamine) is 3-4 times more potent than the *l*-isomer in CNS effects
    - *l*-isomer slightly more potent in cardiovascular actions
  - Mixed amphetamine salts (Adderall®)
    - *d*-isomer to *l*-isomer = 3:1 ratio
  - Racemic mixture (Evekeo®)
- Dextroamphetamine (Dexedrine®)

Katzung's Basic and Clinical Pharm, 16e, 2024 (Ch. 9),
Goodman & Gilman 14e, 2023 (Ch. 14)

Stimulants: Amphetamines

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- Lisdexamfetamine (Vyvanse®)
  - L-lysine, covalently bound to *d*-amphetamine
  - Metabolized by red blood cells to *d*-amphetamine and L-lysine
  - Potential for reduced risk of abuse
- [Methamphetamine (Desoxyn®- former product name)]

Stimulants: Methylphenidate

- MOA: DA and Nor increased in synaptic cleft through blockade of reuptake transporters (non-selective)
- Methylphenidate (Ritalin®)
  - Multiple formulations are available
- Dexmethylphenidate (Focalin®)

Katzung's Basic and Clinical Pharm, 16e, 2024 (Ch. 9), Goodman & Gilman 14e, 2023 (Ch. 14)

Stimulants: Mechanisms of Action

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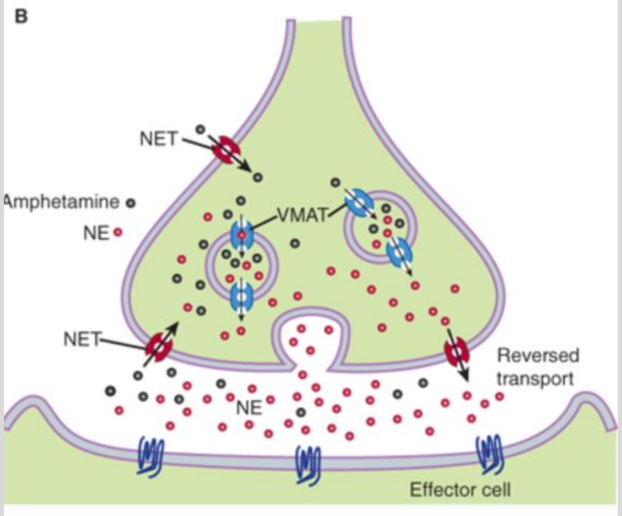
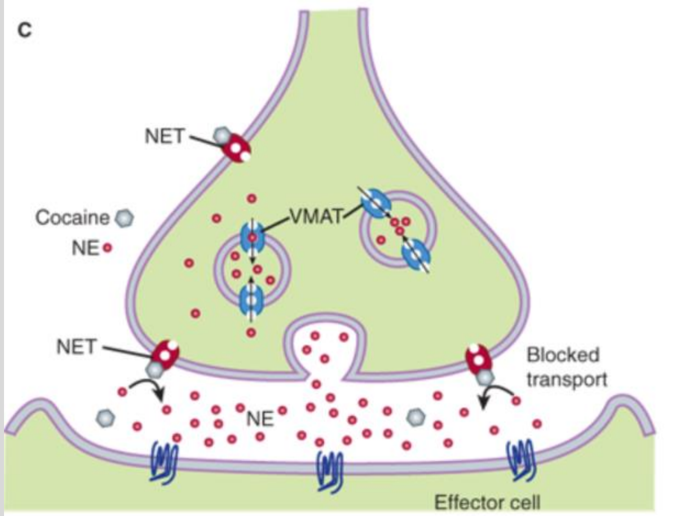


Amphetamine



Methylphenidate

Katzung's Basic and Clinical Pharm, 16e, 2024 (Ch. 9): Adrenoceptor Agonists & Sympathomimetic Drugs (Fig. 9-3)

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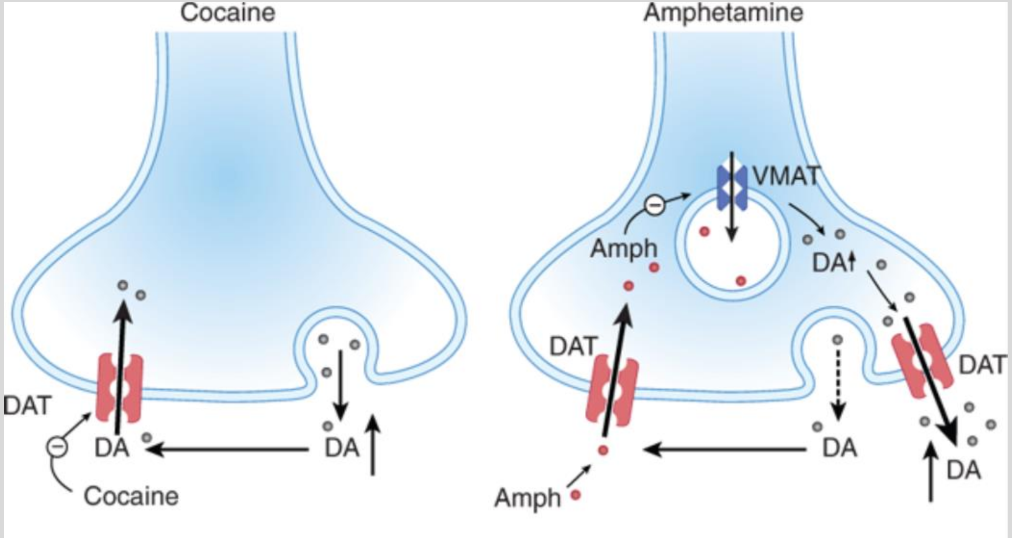

Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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Mechanism of action of cocaine and amphetamine on synaptic terminal of dopamine (DA) neurons. Left: Cocaine inhibits the dopamine transporter (DAT), decreasing DA clearance from the synaptic cleft and causing an increase in extracellular DA concentration. Right: Since amphetamine (Amph) is a substrate of the DAT, it competitively inhibits DA transport. In addition, once in the cell, amphetamine interferes with the vesicular monoamine transporter (VMAT) and impedes the filling of synaptic vesicles. As a consequence, vesicles are depleted and cytoplasmic DA increases. This leads to a reversal of DAT direction, strongly increasing nonvesicular release of DA, and further increasing extracellular DA concentrations.

Citation: Drugs of Abuse, Katzung BG. *Basic & Clinical Pharmacology, 14e;* 2017. Available at:
https://accessmedicine.mhmedical.com/content.aspx?bookid=2249&sectionid=175220606 Accessed: March 20, 2019
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Initiation, Titration & Monitoring of Stimulants

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- Drug therapy should be chosen based on dosing, duration, side effects, patient’s ability to swallow a pill, cost.
- Height, weight, eating and sleeping patterns should be recorded at baseline and every 3 months.
- Initial dose should be titrated to individualize dose and to minimize adverse effects.
  - Can be titrated on a 3-to-7-day basis.
  - Dose may be titrated with a weekly or biweekly telephone call to the family during the first month.
- A face-to face follow-up visit is recommended by the fourth week of medication.
  - Response to medication can be assessed, as well as adverse effects, pulse, blood pressure, and weight.

DePiro JT. 10e, 2017 (Ch. 63) & Pediatrics. 2011;128:1007–1022.

Stimulants: Adverse Effects

- Typically dose-related
- Insomnia
  - Dose stimulants earlier in the day
  - Less insomnia with IR formulations
- Growth suppression
  - Weight loss – minimal for most children
    - Methylphenidate: ↓ 3 kg in first year, 1.2 kg in 2nd year
    - Primarily due to reduced food intake vs ↑ in metabolism
    - Administer high calorie meal at breakfast or dinner; snacks
    - Products vary regarding administration with food (methylphenidate IR products, take 30-45 minutes before a meal)
- Headache - treat
- Abdominal pain
  - Administer with food; make sure meals are eaten

DePiro JT, 10e, 2017 (Ch. 63) & Goodman & Gilman ,14e, 2023 (Ch. 14)

Stimulants: Adverse Effects

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- Cardiovascular effects
  - Average increase in heart rate (1-2 BPM)
  - Increase in blood pressure (1-4 mmHg for SBP and DBP)
  - 5-15% may have more substantial increases in HR and BP

AAP Clinical Practice Guidelines. Pediatrics 2019; 144(4):e2019528.

Stimulants: Adverse Effects

- Baseline screening recommendations
  - Medical and family history
    - Sudden death, CV symptoms, Wolff-Parkinson-White syndrome, hypertrophic cardiomyopathy, long QT syndrome
  - Avoid stimulants if known structural cardiac abnormalities
  - Baseline electrocardiogram if risk factors present

AAP Clinical Practice Guidelines. Pediatrics 2019; 144(4):e2019528

Stimulants: Adverse Effects

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- Zhang L, Honghui Y, Lin L, et al. Risk of cardiovascular diseases associated with medications used in attention-deficit/hyperactivity disorder: A systematic review and meta-analysis. JAMA Network 2022;5(11):e2243597.
- 19 observational studies investigating associations between ADHD medication use and risk of any CVD across age groups
- Outcomes:
  - No significant association with risk of CVD events across age groups
  - No significant association of CVD risk in subgroup analyses
    - 2/8 studies with long-term follow-up showed increased risk in those with history of CVD

JAMA Netw Open. 2022;5(11):e2243597. doi:10.1001/jamanetworkopen.2022.43597

Stimulants: Adverse Effects

- Tics or abnormal movements
  - A tic is a brief, rapid, recurrent, and seemingly purposeless stereotyped motor contraction (blinking, twitching of the nose, jerking of the neck).
- Priapism
  - Definition: prolonged (> 4 hours) and sometimes painful erection
  - Post-marketing cases with methylphenidate prompted warning in December 2013
  - Mean age of patients has been 12.5 years
  - Has occurred on and off the medication

DePiro JT. Pharmacotherapy: A Pathophysiologic Approach., 10e, 2017 (Ch. 63): Attention Deficit/Hyperactivity Disorder.

Stimulants: Efficacy

- Improvement in scholastic performance has been documented in elementary students.
  - Increased reading and math score compared to non-treated patients
  - Scores still lag behind non-ADHD peers
- Scales used at baseline can be evaluation tools for efficacy.
- Conners Comprehensive Behavior Rating Scales (Conners CBRS)
  - One for teacher, parent, child

Shier AC, et al. 2013; 5: 3-11 & DePiro JT, 10e, 2017

Stimulants: Lack of Desired Response

- Increase the dose:
  - Off-label high doses – potential increase in cardiac events
- Variation in treatment response
  - Pharmacology of medications
  - Pharmacokinetic variations
  - Pharmacogenetic differences
- Change to a different formulation or to a different class of stimulant
  - Patient history is important – (ex, medication effective but wearing off)
- Dose conversion information is available for switching between products

DePiro JT. Pharmacotherapy: A Pathophysiologic Approach., 10e, 2017 (Ch. 63): Attention Deficit/Hyperactivity Disorder.

Stimulants: Dependence

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WARNING: ABUSE, MISUSE, AND ADDICTION

Adderall® has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction. Misuse and abuse of CNS stimulants, including Adderall®, can result in overdose and death, and this risk is increased with higher doses or unapproved methods of administration, such as snorting or injection.

Before prescribing Adderall®, assess each patient's risk for abuse, misuse, and addiction. Educate patients and their families about these risks, proper storage of the drug, and proper disposal of any unused drug. Throughout Adderall® treatment, reassess each patient's risk of abuse, misuse, and addiction and frequently monitor for signs and symptoms of abuse, misuse, and addiction.

Adderall® Product Information, accessed 1/9/26

Non-Stimulant Medications

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- Alternatives to stimulants if co-morbidities present
- Not as effective as stimulant medications
- Onset is 2-4 weeks vs 1-2 hours with stimulants
- Patients with cardiovascular risk factors at baseline may need further evaluation (electrocardiogram, consult with a cardiologist)

Pediatrics. 2011;128:1007–1022; Pediatrics 2019; 144(4):e2019528

Nonstimulants: Atomoxetine

- Atomoxetine (Strattera®)
- MOA: Selective inhibitor of Nor reuptake transporter
- Slower therapeutic benefit than stimulants
  - Full benefit may take 6 to 12 weeks
  - May need to overlap with stimulants when transitioning from stimulants because of delayed effect.

Katzung's Basic and Clinical Pharm, 16e, 2024 (Ch. 9) & DePiro JT. 10e, 2017 (Ch. 63)

Nonstimulants: Atomoxetine

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- Adverse effects:
  - Abdominal pain
  - Somnolence
  - Decreased appetite
  - Cardiovascular effects: ↑ in heart rate and blood pressure
- Metabolized by CYP2D6
  - Poor metabolizers have AUCs that are 10x ↑ and peak concentrations that are 4x ↑ extensive metabolizers
- Rare liver injury may occur
  - Routine monitoring of hepatic labs is not recommended
  - Should be discontinued if signs of liver injury present (pruritus, dark urine, jaundice, right upper quadrant tenderness)
- Warning about suicidal ideation due to mechanism that is similar to antidepressants

DePiro JT. Pharmacotherapy: A Pathophysiologic Approach., 10e, 2017 (Ch. 63) Attention Deficit/Hyperactivity Disorder & Strattera® product information

Nonstimulants: Viloxazine

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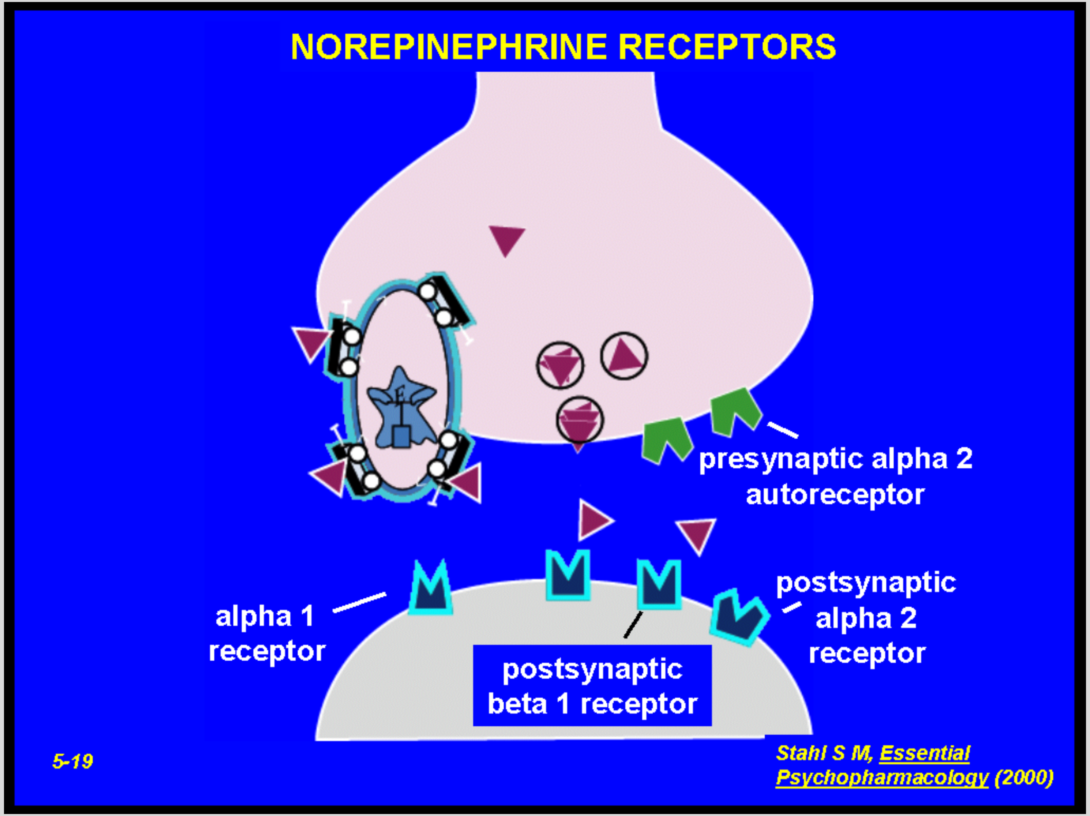
- Viloxazine (Qelbree®)
- MOA: Selective inhibitor of Nor reuptake transporter
- Adverse effects
  - Somnolence
  - Headache
  - Decreased appetite
  - Cardiovascular effects: ↑ in heart rate and blood pressure
- Warning about suicidal ideation

Katzung's Basic and Clinical Pharm, 16e, 2024 (Ch. 9) & DePiro JT. 10e, 2017 (Ch. 63)

Nonstimulants: Alpha 2a Adrenergic Agonists

- MOA: Traditionally defined as *pre*-synaptic alpha-2 agonists (eg, antihypertensive agents); data indicates a *post*-synaptic alpha-2A agonist effects in ADHD
- Post-synaptic effect associated with increased blood flow in prefrontal cortex, which is thought to enhance memory and improve executive functioning.
- Indicated as monotherapy or in combination with stimulants
- Are options for patients who can not tolerate stimulant side effects, or who have co-morbid tics or aggression, high risk for drug dependence
- Guanfacine, extended-release (Intuniv®); can give once daily
- Clonidine, extended-release (Kapvay®)

DePiro JT, 10e, 2017 & Katzung's Basic and Clin Pharm, 16e, 2024 (Ch. 6)



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Nonstimulants: Alpha 2a Adrenergic Agonists

- Adverse effects
- Sedation
  - May be a benefit
- Bradycardia and hypotension
- Potential for rebound hypertension when stopped
- Constipation

DePiro JT. Pharmacotherapy: A Pathophysiologic Approach., 10e, 2017 (Ch. 63): Attention Deficit/Hyperactivity Disorder.

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FDA Warning: Stimulants, Atomoxetine, Alpha-2 Agonists

- Stimulants, atomoxetine, alpha-2 agonists all have a warning about the potential for certain psychiatric adverse effects.
- Psychiatric cases have been reported with all three classes, but more frequently with the stimulants.
- Three categories of effects:
  - Psychosis – hallucinations involving visual or tactile sensations; delusions
  - Mood disturbances – irritability, depression, labile mood
  - Severe anxiety/panic attacks

DePiro JT. Pharmacotherapy: A Pathophysiologic Approach., 10e, 2017 (Ch. 63): Attention Deficit/Hyperactivity Disorder.

Pharmacotherapy: Drug Holiday

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- ‘Holidays’ are periods of time off medication
  - Weekends, summertime
- Potential benefits
  - ‘Catch-up’ in growth
  - Re-assessment for continued need of drug therapy
- Benefit must be balanced against any negative impact of stopping medication, such as learning, self-image, socialization
- Atomoxetine and alpha-2 agonists have a more gradual onset of action and longer duration of effect
  - Drug holidays not recommended

DePiro JT. Pharmacotherapy: A Pathophysiologic Approach., 10e, 2017 (Ch. 63): Attention Deficit/Hyperactivity Disorder

Nonstimulants: Buproprion

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- Bupropion (Wellbutrin®, Zyban®)
  - MOA: A weak Nor and DA reuptake inhibitor
  - Approved for depression, smoking cessation
  - Used off-label in children, adolescents, and adults
  - Considered an alternative to other agents; benefit inconsistent
  - Adverse effects: appetite suppression (less than stimulants), insomnia, tachycardia; may lower seizure threshold in some patients
  - Class suicidal ideation warning

DePiro JT. Pharmacotherapy: A Pathophysiologic Approach., 10e, 2017 (Ch. 63) Attention Deficit/Hyperactivity Disorder; Goodman & Gilman 14e, 2023 (Ch. 18)

Pharmacotherapy of ADHD: Practice Points

- Stimulants are Schedule II drugs per the federal Drug Enforcement Agency (DEA)
  - Prescriptions can not be refilled.
  - Criteria for prescribing stimulants in the treatment of ADHD may vary by state.
- When to stop ADHD medications
  - Individualized discussion with patient and/or parents

Summary Slide: Pharmacological Management of ADHD

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- Pharmacotherapy is only one management modality.
- Therapeutic goals should be identified before medication is initiated.
- Stimulants are first-line therapy pharmacotherapy for most patients.
- Stimulant doses should be titrated to maximize efficacy and minimize adverse effects.
- Patients should be periodically assessed for efficacy and toxicity of ADHD medications.

Summary Slide: Pharmacological Management of ADHD

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Core references:

1. Katzung's Basic and Clinical Pharmacology, 16e, 2024;

   Chapter 6. Introduction to Autonomic Pharmacology. Section "Adrenergic Transmission "

   Chapter 9: Adrenoceptor Agonists & Sympathomimetic Drugs. Sections "The Norepinephrine Transporter" & "Indirect-Acting Sympathomimetics"

2. Reynolds A; Angulo A; Breheney M; Green J; Goldson E. Child Development & Behavior. In: Bunik M, Hay WH, Levin MJ, Abzug MJ, eds. Current Diagnosis & Treatment: Pediatrics, 26e, 2022; Chapter 3, Child Development & Behavior, "Attention-Deficit/Hyperactivity Disorder" section.

- What is a stimulant
(in the management of ADHD)?

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R